Technical specification template: Busbar Trunking (Busduct)

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for busbar trunking (busduct) for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 61439-6 (busbar trunking systems)
- IEC 61439-1 (general rules)
- Local fire barrier and building requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Rated current	[At least the design current after derating; preliminary 4,136 A on the path-out case]	Pending manufacturer data
Rated short-time withstand and peak	[At least the calculated LV fault level]	NOT CALCULATED
Impedance (resistance and reactance per km)	[To be determined by the project]	Pending manufacturer data, needed for voltage drop
Length and routing	15 m (illustrative)	UNVERIFIED_PLACEHOLDER
Ingress protection, material	[Project to select]	Project decision

5. Technical requirements

5.1 The busduct shall be design verified per IEC 61439-6 with evidence supplied.

5.2 Rated current shall account for ambient temperature, altitude and mounting orientation.

5.3 The manufacturer shall declare resistance and reactance per length, expansion provisions, fire barrier arrangements and tap-off units.

5.4 Short-circuit withstand shall exceed the prospective fault level including the peak at the busduct location.

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
losses.distribution_loss_fraction	Distribution losses	Transformer loss report and busduct and cable loss data	UNVERIFIED_PLACEHOLDER
distribution.BD-TX-A.rated_current_a	BD-TX-A: rated_current_a	Busduct data sheet with the manufacturer's type-test evidence	MISSING
distribution.BD-TX-A.resistance_ohm_per_km	BD-TX-A: resistance_ohm_per_km	Busduct data sheet with the manufacturer's type-test evidence	MISSING
distribution.BD-TX-A.reactance_ohm_per_km	BD-TX-A: reactance_ohm_per_km	Busduct data sheet with the manufacturer's type-test evidence	MISSING
distribution.BD-TX-A.short_circuit_withstand_ka	BD-TX-A: short_circuit_withstand_ka	Busduct data sheet with the manufacturer's type-test evidence	MISSING
short_circuit.assumed_clearing_time_s	Assumed fault clearing time for cable and busduct withstand	Protection coordination study result and protective device time-current curves	UNVERIFIED_PLACEHOLDER

7. Inspection and quality requirements

7.1 Review of verification evidence, shop drawings and the ITP.

7.2 Inspection of joints, insulation and supports.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Routine verification per the standard: dielectric test, continuity, mechanical checks.

8.2 Witnessing: [project to define].

8.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Joint torque and insulation resistance, phase sequence.

9.2 Thermography under load.

9.3 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.